Talks and Lectures

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A record of the talks and lecture series I have delivered as a graduate student. All were given at the University of Rochester.

1. Summer Lecture Series: Introduction to Polymer Models

A six-lecture introductory series on directed polymers in a random environment, aimed at graduate students with a background in basic probability. The series developed the model from first principles and built toward the modern picture of the weak-disorder / strong-disorder phase transition.

- **Lecture 1: Directed Polymers in a Random Environment**
  Summer Lecture Series, University of Rochester
  The polymer measure on nearest-neighbour paths, the random environment $\{\omega(n,x)\}$, and the partition function $Z_n = \mathbb{E}_S\big[\exp\big(\beta \sum_{k=1}^{n} \omega(k, S_k)\big)\big]$. Motivation from statistical mechanics and first examples.

- **Lecture 2: The Partition Function as a Martingale**
  Summer Lecture Series, University of Rochester
  The normalised partition function $W_n = Z_n/\mathbb{E}[Z_n]$, its martingale property with respect to the environment filtration, and the almost-sure limit W_∞.

- **Lecture 3: The Zero–One Law and the Phase Transition**
  Summer Lecture Series, University of Rochester
  The dichotomy $\mathbb{P}(W_\infty = 0) \in \{0,1\}$, the definition of the weak- and strong-disorder phases, and the critical inverse temperature β_c.

- **Lecture 4: Weak Disorder and L^2 Techniques**
  Summer Lecture Series, University of Rochester
  Second moment computations, the replica overlap $\sum_n \mathbb{P}(S_n = \tilde{S}_n)$, and uniform integrability of W_n in dimensions $d \geq 3$.

- **Lecture 5: Strong Disorder, Localisation, and Low Dimensions**
  Summer Lecture Series, University of Rochester
  Why strong disorder holds for all $\beta > 0$ when $d = 1, 2$; localisation of the path measure and the role of the free energy.

- **Lecture 6: Scaling Limits and Connections to the KPZ Class**
  Summer Lecture Series, University of Rochester
  Intermediate disorder scaling, convergence of the rescaled partition function to the solution of the stochastic heat equation, and an outlook on KPZ universality.

2. Graduate Student Seminar: The Stochastic Heat Equation

Two talks for the Graduate Student Seminar introducing the stochastic heat equation and its role as a scaling limit in random-environment models.

- **The Stochastic Heat Equation, Part I: Construction and Mild Solutions**
  Graduate Student Seminar, University of Rochester
  Space–time white noise, the multiplicative-noise equation $\partial_t u = \frac{1}{2}\Delta u + \beta\, u\, \dot{W}$, Walsh stochastic integration, and existence and uniqueness of mild solutions via Picard iteration.

- **The Stochastic Heat Equation, Part II: Chaos Expansion and the KPZ Connection**
  Graduate Student Seminar, University of Rochester
  The Wiener chaos expansion of the solution, positivity and moment bounds, the Hopf–Cole transform $h = \log u$ linking SHE to the KPZ equation, and intermittency.

3. Probability Seminar

- **The Second Moment Method for the Partition Function**
  Probability Seminar, University of Rochester
  A talk on the second moment method as a tool for establishing weak disorder. Starting from the Paley–Zygmund inequality $\mathbb{P}(W > 0) \geq (\mathbb{E}W)^2/\mathbb{E}[W^2]$, I computed $\mathbb{E}[W_n^2]$ in terms of the expected overlap of two independent copies of the underlying random walk, showed how transience in $d \geq 3$ makes this second moment bounded uniformly in n, and deduced L^2-boundedness of the martingale W_n and hence $\mathbb{P}(W_\infty > 0) = 1$ for small β. I also discussed where the method fails and what replaces it in low dimensions.

Simple Random Walks and Discrete Stochastic Differential Equations
Vertical Integration Workshop, University of Rochester April 18, 2026
We study simple random walk models in random environments and their connection to stochastic partial differential equations. We discuss how suitably scaled partition functions can be interpreted as discrete analogues of solutions to the Parabolic Anderson Model (PAM). This perspective is a natural framework for constructing SPDE solutions via discrete models. Joint work with Arjun Krishnan.
